Module 1 – MAT2002– Discrete Mathematics and Graph Theory

Predicate Calculus

- Introduction to logic and predicate calculus
- Statements and propositions
- Logical notation and symbols
- Atomic and compound statements
- Logical connectives
- Truth tables
- Tautologies and contradictions
- Logical equivalence
- Implications and converse statements
- Laws of statement calculus
- Rules of inference
- Predicate logic
- Predicates and quantified statements
- Universal and existential quantifiers
- Nested quantifiers
- Rules of inference for predicates
- Rules of inference for quantified statements
- Applications in computer science

Learning Objectives

- Understand the foundations of mathematical logic and predicate calculus.
- Learn the language and notation used in propositional and predicate logic.
- Construct and analyze compound logical statements using connectives.
- Develop truth tables for logical propositions.
- Identify tautologies, contradictions and contingencies.
- Apply laws of logical equivalence in simplifying propositions.
- Understand implications and logical consequence.
- Apply rules of inference in formal reasoning.
- Study predicates and quantified statements rigorously.
- Understand the meaning and applications of universal and existential quantifiers.
- Analyze statements involving nested quantifiers.
- Apply logical reasoning in computer science, databases and artificial intelligence.

1 Introduction to Predicate Calculus

Definition 1. Logic is the branch of mathematics that deals with principles of valid reasoning and inference.

Note 1. Predicate calculus extends propositional logic by introducing variables, predicates and quantifiers.

1.1 Applications of Logic

- Artificial Intelligence
- Database Systems
- Computer Programming
- Compiler Design
- Digital Circuit Design
- Formal Verification
- Cryptography

2 Statements and Propositions

Definition 2. A statement or proposition is a declarative sentence that is either true or false, but not both.

Example 1. "7 is a prime number" is a proposition.

Example 2. "The Earth revolves around the Sun" is a proposition.

Example 3. "x+5>7" is not a proposition unless the value of x is specified.

2.1 Types of Statements

- Atomic statements
- Compound statements

Definition 3. A statement that cannot be divided further is called an atomic statement.

Definition 4. Statements formed using logical connectives are called compound statements.

3 Logical Connectives

Symbol	Name	Meaning
$\neg p$	Negation	Not p
$p \wedge q$	Conjunction	p and q
$p \vee q$	Disjunction	p or q
$p \rightarrow q$	Implication	If p then q
$p \leftrightarrow q$	Biconditional	p iff q

4 Order of Precedence for Logical Connectives

We will generally use parentheses to specify the order in which logical operators in a compound proposition are to be applied. For example,

$$(p \vee q) \wedge (\neg r)$$

is the conjunction of $p \vee q$ and $\neg r$. However, to avoid the use of an excessive number of parentheses, we adopt an order of precedence for the logical operators, given as follows:

1. The negation operator has precedence over all other logical operators.

   Thus
   $$\neg p \wedge q$$
   means
   $$(\neg p) \wedge q$$
   not
   $$\neg(p \wedge q)$$

2. The conjunction operator has precedence over the disjunction operator.

   Thus
   $$p \wedge q \vee r$$
   means
   $$(p \wedge q) \vee r$$
   but not
   $$p \wedge (q \vee r)$$

3. The conditional and biconditional operators
   $$\rightarrow \quad \text{and} \quad \leftrightarrow$$
   have lower precedence than other operators. Among them,
   $$\rightarrow$$
   has precedence over
   $$\leftrightarrow$$

4.1 Truth Table for Negation

p	$\neg p$
T	F
F	T

4.2 Truth Table for Conjunction

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

4.3 Truth Table for Disjunction

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Example 4. Construct a truth table for each of the following compound propositions:

1. $$(p \vee q) \rightarrow (p \wedge q)$$

2. $$(p \rightarrow q) \rightarrow (q \rightarrow p)$$

3. $$(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$$

4. $$(p \leftrightarrow q) \leftrightarrow ((p \wedge q) \vee (\neg p \wedge \neg q))$$

5. $$(\neg p \leftrightarrow \neg q) \leftrightarrow (p \leftrightarrow q)$$

Solution (1)

p	q	$p \vee q$	$p \wedge q$	$(p \vee q) \rightarrow (p \wedge q)$
T	T	T	T	T
T	F	T	F	F
F	T	T	F	F
F	F	F	F	T

Solution (2)

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \rightarrow (q \rightarrow p)$
T	T	T	T	T
T	F	F	T	T
F	T	T	F	F
F	F	T	T	T

Solution (3)

p	q	$\neg p$	$q \to \neg p$	$p \leftrightarrow q$	$(q \to \neg p) \leftrightarrow (p \leftrightarrow q)$
T	T	F	F	T	F
T	F	F	T	F	F
F	T	T	T	F	F
F	F	T	T	T	T

Solution (4)

p	q	$p \leftrightarrow q$	$p \wedge q$	$\neg p$	$\neg q$	$(p \wedge q) \vee (\neg p \wedge \neg q)$
T	T	T	T	F	F	T
T	F	F	F	F	T	F
F	T	F	F	T	F	F
F	F	T	F	T	T	T

$(p \leftrightarrow q) \leftrightarrow ((p \wedge q) \vee (\neg p \wedge \neg q))$
T
T
T
T

Solution (5)

p	q	$\neg p$	$\neg q$	$\neg p \leftrightarrow \neg q$	$p \leftrightarrow q$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

$(\neg p \leftrightarrow \neg q) \leftrightarrow (p \leftrightarrow q)$
T
T
T
T

Example 5. Construct the truth table for each of the compound propositions given as follows:

1. $$((p \to (q \to r)) \to ((p \to q) \to (p \to r)))$$

2. $$\neg(p \vee (q \wedge r)) \leftrightarrow ((p \vee q) \wedge (p \to r))$$

3. $$(\neg p \leftrightarrow \neg q) \leftrightarrow (q \leftrightarrow r)$$

4.
$$(p \to (q \to s)) \land (\neg r \lor p) \land q$$

5.
$$((p \to q) \to r) \to s$$

5 Tautologies and Contradictions

Definition 5. A compound proposition that is always true is called a tautology.

Example 6.
$$p \lor \neg p$$
is a tautology.

Definition 6. A compound proposition that is always false is called a contradiction.

Example 7.
$$p \land \neg p$$
is a contradiction.

Example 8. Determine which of the following compound propositions are tautologies and which of them are contradictions, using truth tables:

1.
$$(\neg q \land (p \to q)) \to \neg p$$

2.
$$((p \to q) \land (q \to r)) \to (p \to r)$$

3.
$$\neg(q \to r) \lor r \land (p \to q)$$

4.
$$((p \lor q) \land (p \to r) \land (q \to r)) \to r$$

Solution (a)

p	q	$p \to q$	$\neg q$	$\neg q \land (p \to q)$	$(\neg q \land (p \to q)) \to \neg p$
T	T	T	F	F	T
T	F	F	T	F	T
F	T	T	F	F	T
F	F	T	T	T	T

$$\therefore (\neg q \land (p \to q)) \to \neg p$$
is a tautology.

Solution (b)

p	q	r	$p \to q$	$q \to r$	$(p \to q) \wedge (q \to r)$	$((p \to q) \wedge (q \to r)) \to (p \to r)$
T	T	T	T	T	T	T
T	T	F	T	F	F	T
T	F	T	F	T	F	T
T	F	F	F	T	F	T
F	T	T	T	T	T	T
F	T	F	T	F	F	T
F	F	T	T	T	T	T
F	F	F	T	T	T	T

$$\therefore ((p \to q) \wedge (q \to r)) \to (p \to r)$$

is a tautology.

Solution (c)

p	q	r	$q \to r$	$\neg(q \to r)$	$p \to q$	$\neg(q \to r) \vee (r \wedge (p \to q))$
T	T	T	T	F	T	T
T	T	F	F	T	T	T
T	F	T	T	F	F	F
T	F	F	T	F	F	F
F	T	T	T	F	T	T
F	T	F	F	T	T	T
F	F	T	T	F	T	T
F	F	F	T	F	T	F

$$\therefore \neg(q \to r) \vee (r \wedge (p \to q))$$

is neither a tautology nor a contradiction.

Solution (d)

p	q	r	$p \vee q$	$p \to r$	$q \to r$	$(p \vee q) \wedge (p \to r) \wedge (q \to r)$	$((p \vee q) \wedge (p \to r) \wedge (q \to r)) \to r$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	T	T	T	T	T
T	F	F	T	F	T	F	T
F	T	T	T	T	T	T	T
F	T	F	T	T	F	F	T
F	F	T	F	T	T	F	T
F	F	F	F	T	T	F	T

$$\therefore ((p \vee q) \wedge (p \to r) \wedge (q \to r)) \to r$$

is a tautology.

6 Logical Equivalence

Definition 7. Two propositions are logically equivalent if they have identical truth values for all possible cases.

6.1 Logical Equivalences Involving Implications

1. $$p \to q \equiv \neg p \vee q$$
2. $$p \to q \equiv \neg q \to \neg p$$
3. $$p \vee q \equiv \neg p \to q$$
4. $$p \wedge q \equiv \neg(p \to \neg q)$$
5. $$\neg(p \to q) \equiv p \wedge \neg q$$
6. $$(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$$
7. $$(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$$
8. $$(p \to q) \vee (p \to r) \equiv p \to (q \vee r)$$
9. $$(p \to r) \vee (q \to r) \equiv (p \wedge q) \to r$$

6.2 Logical Equivalences Involving Biconditionals

1. $$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$$
2. $$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$
3. $$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$
4. $$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

7 Laws of Algebra of Propositions

Sl. No.	*Name of the law*	*Primal form*	*Dual form*
1.	Idempotent law	$p \vee p \equiv p$	$p \wedge p \equiv p$
2.	Identity law	$p \vee F \equiv p$	$p \wedge T \equiv p$
3.	Dominant law	$p \vee T \equiv T$	$p \wedge F \equiv F$
4.	Complement law	$p \vee \neg p \equiv T$	$p \wedge \neg p \equiv F$
5.	Commutative law	$p \vee q \equiv q \vee p$	$p \wedge q \equiv q \wedge p$
6.	Associative law	$(p \vee q) \vee r \equiv p \vee (q \vee r)$	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
7.	Distributive law	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
8.	Absorption law	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
9.	De Morgan's law	$\neg(p \vee q) \equiv \neg p \wedge \neg q$	$\neg(p \wedge q) \equiv \neg p \vee \neg q$

Example 9. Without using truth tables, prove the following:

1. $$(\neg p \vee q) \wedge (p \wedge (p \wedge q)) \equiv p \wedge q$$

2. $$p \to (q \to p) \equiv \neg p \to (p \to q)$$

3. $$\neg(p \leftrightarrow q) \equiv (p \vee q) \wedge \neg(p \vee q) \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$$

Solution (1)

$$(\neg p \vee q) \wedge (p \wedge (p \wedge q))$$

Using Associative Law,

$$\equiv (\neg p \vee q) \wedge ((p \wedge p) \wedge q)$$

Using Idempotent Law,

$$\equiv (\neg p \vee q) \wedge (p \wedge q)$$

Using Commutative Law,

$$\equiv (p \wedge q) \wedge (\neg p \vee q)$$

Using Distributive Law,

$$\equiv (p \wedge q \wedge \neg p) \vee (p \wedge q \wedge q)$$

Using Complement Law,

$$\equiv (F \wedge q) \vee (p \wedge q)$$

Using Dominant Law,

$$\equiv F \vee (p \wedge q)$$

Using Identity Law,

$$\equiv p \wedge q$$

Hence proved.

Solution (2)

$$p \to (q \to p)$$

Using implication law,

$$\equiv \neg p \vee (\neg q \vee p)$$

Using Associative Law,

$$\equiv (\neg p \vee p) \vee \neg q$$

Using Complement Law,

$$\equiv T \vee \neg q$$

Using Dominant Law,

$$\equiv T$$

Now consider,

$$\neg p \to (p \to q)$$

Using implication law,

$$\equiv p \vee (\neg p \vee q)$$

Using Associative Law,

$$\equiv (p \vee \neg p) \vee q$$

Using Complement Law,

$$\equiv T \vee q$$

Using Dominant Law,

$$\equiv T$$

Therefore,

$$p \to (q \to p) \equiv \neg p \to (p \to q)$$

Hence proved.

Solution (3)

$$\neg(p \leftrightarrow q)$$

Using biconditional equivalence,

$$\equiv \neg\left[(p \wedge q) \vee (\neg p \wedge \neg q)\right]$$

Using De Morgan's Law,

$$\equiv \neg(p \wedge q) \wedge \neg(\neg p \wedge \neg q)$$

Again using De Morgan's Law,

$$\equiv (\neg p \vee \neg q) \wedge (p \vee q)$$

Using Distributive Law,

$$\equiv (\neg p \wedge p) \vee (\neg p \wedge q) \vee (\neg q \wedge p) \vee (\neg q \wedge q)$$

Using Complement Law,

$$\equiv F \vee (\neg p \wedge q) \vee (p \wedge \neg q) \vee F$$

Using Identity Law,

$$\equiv (\neg p \wedge q) \vee (p \wedge \neg q)$$

Using Commutative Law,

$$\equiv (p \wedge \neg q) \vee (\neg p \wedge q)$$

Hence proved.

Example 10. Without constructing the truth tables, prove the following:

1. $$\neg p \to (q \to r) \equiv q \to (p \vee r)$$

2. $$p \to (q \to r) \equiv p \to (\neg q \vee r) \equiv (p \wedge q) \to r$$

3. $$((p \vee q) \wedge \neg(\neg p \wedge (\neg q \vee \neg r))) \vee (\neg p \wedge \neg q) \vee (\neg p \wedge \neg r)$$

   is a tautology.

8 Theory of Inference

1.

$$(p \wedge q) \rightarrow p$$

$$(p \wedge q) \rightarrow q$$

(Simplification)

2.

$$p \rightarrow (p \vee q)$$

$$q \rightarrow (p \vee q)$$

(Addition)

3.

$$(p) \wedge (q) \rightarrow (p \wedge q)$$

(Conjunction)

4.

$$[p \wedge (p \rightarrow q)] \rightarrow q$$

(Modus Ponens)

5.

$$[\neg q \wedge (p \rightarrow q)] \rightarrow \neg p$$

(Modus Tollens)

6.

$$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$$

(Hypothetical Syllogism)

7.

$$[(p \vee q) \wedge \neg p] \rightarrow q$$

(Disjunctive Syllogism)

8.

$$[(p \vee q) \wedge (\neg p \vee r)] \rightarrow (q \vee r)$$

(Resolution)

Example 11. Show that

$$(t \wedge s)$$

can be derived from the premises

$$p \to q, \quad q \to \neg r, \quad r, \quad p \vee (t \wedge s).$$

Step No.	*Statement*	*Reason*
1.	$p \to q$	P
2.	$q \to \neg r$	P
3.	$p \to \neg r$	1, 2, Hypothetical syllogism
4.	$r \to \neg p$	3, and $p \to q \equiv \neg q \to \neg p$
5.	r	P
6.	$\neg p$	4, 5, and Modus ponens
7.	$p \vee (t \wedge s)$	P
8.	$t \wedge s$	6, 7, and Disjunctive syllogism

Example 12. Show that

$$(a \vee b)$$

follows logically from the premises

$$p \vee q, \quad (p \vee q) \to \neg r, \quad \neg r \to (s \wedge \neg t)$$

and

$$(s \wedge \neg t) \to (a \vee b).$$

Step No.	*Statement*	*Reason*
1.	$(p \vee q) \to \neg r$	P
2.	$\neg r \to (s \wedge \neg t)$	P
3.	$(p \vee q) \to (s \wedge \neg t)$	1, 2, and Hypothetical syllogism
4.	$p \vee q$	P
5.	$s \wedge \neg t$	3, 4, and Modus ponens
6.	$(s \wedge \neg t) \to (a \vee b)$	P
7.	$a \vee b$	5, 6, and Modus ponens

Example 13. Show that

$$(p \to q) \wedge (r \to s), \quad (q \to t) \wedge (s \to u), \quad \neg(t \wedge u)$$

and

$$(p \to r) \Rightarrow \neg p.$$

Step No.	*Statement*	*Reason*
1.	$(p \to q) \wedge (r \to s)$	P
2.	$p \to q$	1, and simplification
3.	$r \to s$	1, and simplification
4.	$(q \to t) \wedge (s \to u)$	P
5.	$q \to t$	4, and simplification
6.	$s \to u$	4, and simplification
7.	$p \to t$	2, 5, and hypothetical syllogism
8.	$r \to u$	3, 6, and hypothetical syllogism
9.	$p \to r$	P
10.	$p \to u$	8, 9, and hypothetical syllogism
11.	$\neg t \to \neg p$	7, and $a \to b \equiv \neg b \to \neg a$
12.	$\neg u \to \neg p$	10, and $a \to b \equiv \neg b \to \neg a$
13.	$(\neg t \vee \neg u) \to \neg p$	11, 12, and $(a \to b),\ (c \to b) \Rightarrow (a \vee c) \to b$
14.	$\neg(t \wedge u) \to \neg p$	13, and De Morgan's law
15.	$\neg(t \wedge u)$	P
16.	$\neg p$	14, 15, and Modus ponens

Example 14. Show that the following set of premises is inconsistent:

If Rama gets his degree, he will go for a job.

If he goes for a job, he will get married soon.

If he goes for higher study, he will not get married.

Rama gets his degree and goes for higher study.

Let the statements be symbolised as follows:

p : Rama gets his degree.

q : He will go for a job.

r : He will get married soon.

s : He goes for higher study.

Then we have to prove that

$$p \to q, \quad q \to r, \quad s \to \neg r, \quad p \wedge s$$

are inconsistent.

Step No.	*Statement*	*Reason*
1.	$p \to q$	P
2.	$q \to r$	P
3.	$p \to r$	1, 2, and hypothetical syllogism
4.	$p \wedge s$	P
5.	p	4, and simplification
6.	s	4, and simplification
7.	$s \to \neg r$	P
8.	$\neg r$	6, 7, and modus ponens
9.	r	3, 5, and modus ponens
10.	$r \wedge \neg r$	8, 9, and conjunction
11.	F	10, and negation law

Hence the set of given premises is inconsistent.

Example 15. If

$$L(x, y)$$

symbolises the statement

$$\text{“}x \text{ loves } y\text{”},$$

where the universe of discourse for both x and y consists of all people in the world, translate the following English sentences into logical expressions:

1. Every body loves z.
2. Every body loves somebody.
3. There is somebody whom everybody loves.
4. Nobody loves everybody.
5. There is somebody whom no one loves.

Solution

1. $$L(x, z)$$

   for all x.

Hence

$$\forall x\ L(x, z)$$

2. $$L(x, y)$$

   is true for all x and some y.

   Hence

   $$\forall x \exists y\ L(x, y)$$

3. Even though, (c) is the same as (b), the stress is on the existence of somebody (y) whom all x love.

   Hence

   $$\exists y \forall x\ L(x, y)$$

4. Nobody loves every body

   i.e., There is not one who loves everybody

   Hence

   $$\neg \exists x \forall y\ L(x, y)$$

   $$\equiv \forall x \neg \forall y\ L(x, y)$$

   $$\equiv \forall x \exists y \neg L(x, y)$$

5. The sentence means that there is somebody whom every one does not love.

   Hence

   $$\neg \forall x \exists y\ L(x, y)$$

   $$\equiv \exists x \neg \exists y\ L(x, y)$$

   $$\equiv \exists x \forall y \neg L(x, y)$$

9 Predicate Logic

Definition 8. A predicate is a statement involving variables that becomes a proposition when specific values are assigned.

Example 16.

$$P(x) : x > 5$$

10 Quantifiers

Definition 9. The universal quantifier is denoted by

$$\forall$$

and means "for all".

Example 17.

$$\forall x \in \mathbb{N},\ x + 1 > x$$

Examples

1. If
$$P(x) \equiv ((-x)^2 = x^2)$$
where the universe consists of all integers, then the truth value of
$$\forall x \left((-x)^2 = x^2\right)$$
is T.

2. If
$$Q(x) \equiv \text{“}2x > x''$$
where the universe consists of all real numbers, then the truth value of
$$\forall x\ Q(x)$$
is F.

3. If
$$P(x) \equiv \text{“}x^2 < 10''$$
where the universe consists of the positive integers
$$1,\ 2,\ 3 \text{ and } 4,$$
then
$$\forall x\ P(x) = P(1) \wedge P(2) \wedge P(3) \wedge P(4)$$
and so the truth value of
$$\forall x\ P(x) = T \wedge T \wedge T \wedge F = F.$$

4. Let
$$P(x) \equiv x \text{ is an integer}$$
and
$$Q(x) \equiv x \text{ is either positive or negative.}$$
Then
$$P(x) \to Q(x)$$

is a compound propositional function.

Obviously

$$\forall x(P(x) \to Q(x)),$$

where the universe of discourse consists of integers.

5. Let

$$P(x, y) : x \text{ is taller than } y.$$

If x is taller than y, then y is not taller than x.

viz.

$$P(x, y) \to \neg P(y, x)$$

Definition 10. The existential quantifier is denoted by

$$\exists$$

and means "there exists".

Example 18.

$$\exists x \in \mathbb{R} : x^2 = 4$$

Examples

1. When

$$P(x)$$

denotes the propositional function

$$\text{“}x > 3\text{”},$$

the truth value of

$$\exists x P(x)$$

is T, where the universe of discourse consists of all real numbers, since

$$\text{“}x > 3\text{”}$$

is true for

$$x = 4.$$

2. When

$$P(x)$$

denotes

$$\text{“}x^2 > 10\text{”},$$

where the universe of discourse consists of the positive integers not exceeding 4, then the truth value of

$$\exists x P(x)$$

is T, since

$$P(1) \vee P(2) \vee P(3) \vee P(4)$$

is true as

$$P(4)$$

[viz.,

$$4^2 > 10$$

] is true.

11 Nested Quantifiers

There are situations when quantifiers occur in combinations in respect of 1-place or n-place predicate formulas (i.e., propositional functions containing 1 or n variables). For example let us consider a 2-place predicate formula

$$P(x, y).$$

Now

$$\forall x\ \forall y\ P(x, y) \equiv \forall x[\forall y\ P(x, y)]$$

$$\equiv \forall y[\forall x\ P(x, y)] \tag{1}$$

and

$$\exists x\ \exists y\ P(x, y) \equiv \exists x[\exists y\ P(x, y)]$$

$$\equiv \exists y[\exists x\ P(x, y)] \tag{2}$$

From the meaning of quantifiers and by (1) and (2) the following simplifications hold good:

$$\forall x\ \forall y\ P(x, y) \Rightarrow (\exists y)\ \forall x\ P(x, y) \Rightarrow \forall x\ \exists y\ P(x, y)$$

$$\forall y\ \forall x\ P(x, y) \Rightarrow (\exists x)\ \forall y\ P(x, y) \Rightarrow \forall y\ \exists x\ P(x, y)$$

Note 2. The negation of multiply quantified predicate formulas may be obtained by applying the rules for negation (given earlier) from left to right.

Thus

$$\neg[\forall x\ \exists y\ P(x, y)] \equiv \exists x[\neg\exists y\ P(x, y)]$$

$$\equiv \exists x\ \forall y[\neg P(x, y)]$$

Example 19.

$$\forall x \exists y(x + y = 0)$$

Note 3. The order of quantifiers is important in predicate logic.

12 Rules of Inference for Quantified Statements

Theorem 1 (Universal Specification (US)). Universal Specification is the rule of inference which states that one can conclude that

$$P(c)$$

is true, if

$$\forall x\ P(x)$$

is true, where c is an arbitrary member of the universe of discourse.

This rule is also called the *universal instantiation*.

Theorem 2 (Existential Specification (ES)). Existential Specification is the rule which allows us to conclude that

$$P(c)$$

is true, if

$$\exists x\ P(x)$$

is true, where c is not an arbitrary member of the universe, but one for which

$$P(c)$$

is true.

Usually we will not know what c is but know that it exists. Since it exists, we may call it c.

This rule is also called the *existential instantiation*.

Theorem 3 (Universal Generalisation (UG)). Universal Generalisation is the rule which states that

$$\forall x\ P(x)$$

is true, if

$$P(c)$$

is true, where c is an arbitrary member (not a specific member) of the universe of discourse.

Theorem 4 (Existential Generalisation (EG)). Existential Generalisation is the rule that is used to conclude that

$$\exists x\ P(x)$$

is true when

$$P(c)$$

is true, where c is a particular member of the universe of discourse.

Examples

1. Let us consider the following "Famous Socrates argument" which is given by:

   All men are mortal.
   Socrates is a man.
   Therefore Socrates is a mortal.

Let us use the notations

$$H(x) : x \text{ is a man}$$

$$M(x) : x \text{ is a mortal}$$

$$s : \text{ Socrates}$$

With these symbolic notations, the problem becomes

$$\forall x(H(x) \to M(x)) \wedge H(s) \Rightarrow M(s)$$

The derivation of the proof is as follows:

Step No.	*Statement*	*Reason*
1.	$\forall x(H(x) \to M(x))$	P
2.	$H(s) \to M(s)$	US, 1
3.	$H(s)$	P
4.	$M(s)$	2, 3, Modus Ponens

2. Application of any of US, ES, UG and EG rules wrongly may lead to a false conclusion from a true premise as in the following example.

   Let

   $$D(u, v) : u \text{ is divisible by } v,$$

   where the universe of discourse is

   $$(5, 6, 10, 11).$$

   Then

   $$\exists u \; D(u, 5)$$

   is true, since

   $$D(5, 5)$$

   and

   $$D(10, 5)$$

   are true.

   But

   $$\forall u \; D(u, 5)$$

is false, since

$$D(6,5)$$

and

$$D(11,5)$$

are false.

We now give the following derivation:

Step No.	*Statement*	*Reason*
1.	$\exists u\ D(u,5)$	P
2.	$D(c,5)$	ES, 1
3.	$\forall x\ D(x,5)$	UG, 2

13 Normal Forms in Propositional Logic

What Are Normal Forms?

Turning any complex logical statement into a standard "recipe" so everyone – or every algorithm – follows the same steps every time.

Why Normal Forms Matter

- Provide a uniform structure for simplification and mechanical reasoning.
- Underpin algorithms like SAT-solvers and automated proof engines.
- Help us to see patterns and apply general rules systematically.

13.1 Conjunctive Normal Form (CNF)

When we are interested in listing all the ways a formula can avoid being false.

Definition 11. Conjunctive Normal Form (CNF) is an "AND of OR-clauses."

- Think of a checklist:

  (item A OR item B) AND (item C OR item D OR item E)

- Example:

  $$(p \vee \neg q) \wedge (r \vee s \vee \neg t)$$

13.2 Disjunctive Normal Form (DNF)

When we are interested in transforming a proposition into a logically equivalent expression that explicitly lists all the ways the formula can be true.

Definition 12. Disjunctive Normal Form (DNF) is an "OR of AND-clauses."

- Think of menu combos:

$$\text{(dish 1 AND side A) OR (dish 2 AND side B)}$$

- Example:

$$(p \wedge q) \vee (\neg p \wedge r \wedge s)$$

13.3 Truth Table Mapping

CNF: Built from rows where formula is false. Each row gives a clause that must be false to falsify the formula.

DNF: Built from rows where formula is true. Each row gives a term that must be true to satisfy the formula.

Example: Formula

$$p \to q$$

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

CNF: From row 2

$$(\neg p \vee q)$$

DNF: From rows 1, 3, 4

$$(p \wedge q) \vee (\neg p \wedge q) \vee (\neg p \wedge \neg q)$$

14 Procedure to Obtain the DNF or CNF of a Given Formula

Step 1

If the connectives

$$\to \text{ and } \leftrightarrow$$

are present in the given formula they are replaced by

$$\wedge,\ \vee \text{ and } \neg$$

viz.

$$p \to q$$

is replaced by

$$\neg p \vee q$$

and

$$p \leftrightarrow q$$

is replaced by either

$$(p \wedge q) \vee (\neg p \wedge \neg q)$$

or

$$(\neg p \vee q) \wedge (\neg q \vee p).$$

Step 2

If the negation is present before the given formula or a part of the given formula (not a variable), De Morgan's laws are applied so that the negation is brought before the variables only.

Step 3

If necessary, the distributive law and the idempotent law are applied.

Step 4

If there is an elementary product which is equivalent to the truth value

$$F$$

in the DNF, it is omitted. Similarly if there is an elementary sum which is equivalent to the truth value

$$T$$

in the CNF, it is omitted.

For example, the DNF of

$$q \rightarrow (q \rightarrow p)$$

is given by

$$q \rightarrow (q \rightarrow p) \equiv \neg q \vee (q \rightarrow p)$$

$$\equiv \neg q \vee (\neg q \vee p)$$

$$\equiv (\neg q \vee \neg q) \vee p, \quad \text{by associative law}$$

$$\equiv \neg q \vee p, \quad \text{by idempotent law.}$$

The CNF of

$$\neg(p \vee q) \leftrightarrow (p \wedge q)$$

is given by

$$\neg(p \vee q) \leftrightarrow (p \wedge q)$$

$$\equiv (\neg(p \vee q) \wedge (p \wedge q)) \vee (\neg[\neg(p \vee q)] \wedge \neg(p \wedge q))$$

$$\equiv (\neg p \wedge \neg q) \wedge (p \wedge q) \vee (p \vee q) \wedge (\neg p \vee \neg q)$$

$$\equiv (p \wedge \neg p) \wedge (q \wedge \neg q) \vee (p \vee q) \wedge (\neg p \vee \neg q)$$

$$\equiv F \wedge F \vee (p \vee q) \wedge (\neg p \vee \neg q)$$

$$\equiv (p \vee q) \wedge (\neg p \vee \neg q)$$

Example 20. Find the disjunctive normal forms of the following statements:

1. $$\neg((p \leftrightarrow q) \wedge r)$$
2. $$p \vee (\neg p \rightarrow (q \vee (q \rightarrow \neg r)))$$
3. $$p \wedge \neg(q \wedge r) \vee (p \rightarrow q)$$
4. $$(p \wedge \neg(q \vee r)) \vee (((p \wedge q) \vee \neg r) \wedge p)$$

Solution (1)

$$\neg((p \leftrightarrow q) \wedge r)$$

$$\equiv \neg[((p \wedge q) \vee (\neg p \wedge \neg q)) \wedge r]$$

$$\equiv \neg[((p \wedge q) \wedge r) \vee ((\neg p \wedge \neg q) \wedge r)]$$

$$\equiv \neg[((p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge r))]$$

$$\equiv \neg(p \wedge q \wedge r) \wedge \neg(\neg p \wedge \neg q \wedge r)$$

$$\equiv (\neg p \vee \neg q \vee \neg r) \wedge (p \vee q \vee \neg r)$$

Solution (2)

$$p \vee (\neg p \rightarrow (q \vee (q \rightarrow \neg r)))$$

$$\equiv p \vee (p \vee (q \vee (\neg q \vee \neg r)))$$

$$\equiv p \vee p \vee q \vee \neg q \vee \neg r$$

$$\equiv p \vee q \vee \neg q \vee \neg r$$

$$\equiv p \vee T \vee \neg r$$

$$\equiv T$$

Note 4. The given statement is a tautology, as

$$p \vee (q \vee \neg q) \vee \neg r \equiv p \vee T \vee \neg r \equiv T$$

Solution (3)

$$p \wedge \neg(q \wedge r) \vee (p \rightarrow q)$$

$$\equiv p \wedge (\neg q \vee \neg r) \vee (\neg p \vee q)$$

$$\equiv (p \wedge \neg q) \vee (p \wedge \neg r) \vee (\neg p \vee q)$$

Solution (4)

$$(p \wedge \neg(q \vee r)) \vee (((p \wedge q) \vee \neg r) \wedge p)$$

$$\equiv (p \wedge \neg q \wedge \neg r) \vee ((p \wedge q \wedge p) \vee (\neg r \wedge p))$$

$$\equiv (p \wedge \neg q \wedge \neg r) \vee (p \wedge q) \vee (p \wedge \neg r)$$

Example 21. Find the conjunction normal forms of the following statements:

1. $$(p \wedge \neg(q \wedge r)) \vee (p \rightarrow q)$$
2. $$(q \vee (p \wedge q)) \wedge \neg((p \vee r) \wedge q)$$
3. $$(p \wedge \neg(q \vee r)) \vee (((p \wedge q) \vee \neg r) \wedge p)$$

Solution (1)

$$(p \wedge \neg(q \wedge r)) \vee (p \rightarrow q)$$

$$\equiv (p \wedge (\neg q \vee \neg r)) \vee (\neg p \vee q)$$

$$\equiv (p \wedge \neg q) \vee (p \wedge \neg r) \vee (\neg p \vee q)$$

$$\equiv (p \vee \neg p \vee q) \wedge (\neg q \vee \neg p \vee q) \wedge (p \vee \neg r \vee \neg p \vee q)$$

$$\equiv T \wedge T \wedge (p \vee \neg r \vee q)$$

$$\equiv p \vee \neg r \vee q$$

Solution (2)

$$[q \vee (p \wedge q)] \wedge \neg[(p \vee r) \wedge q]$$

$$\equiv q \wedge \neg[(p \vee r) \wedge q], \quad \text{by absorption law}$$

$$\equiv q \wedge [\neg(p \vee r) \vee \neg q]$$

$$\equiv q \wedge [(\neg p \wedge \neg r) \vee \neg q]$$

$$\equiv q \wedge (\neg p \vee \neg q) \wedge (\neg q \vee \neg r)$$

Solution (3)

$$(p \wedge \neg(q \vee r)) \vee (((p \wedge q) \vee \neg r) \wedge p)$$

$$\equiv (p \wedge \neg q \wedge \neg r) \vee ((p \vee \neg r) \wedge (q \vee \neg r) \wedge p)$$

$$\equiv (p \wedge \neg q \wedge \neg r) \vee (p \wedge (q \vee \neg r)), \quad \text{by absorption law}$$

$$\equiv [(p \wedge \neg q \wedge \neg r) \vee p] \wedge [(p \wedge \neg q \wedge \neg r) \vee (q \vee \neg r)]$$

$$\equiv p \wedge [(p \wedge \neg q \wedge \neg r) \vee q], \quad \text{by absorption law}$$

$$\equiv p \wedge (q \vee \neg r), \quad \text{by absorption law}$$

15 Principal Normal Form

What Are Principal Normal Forms?

Principal normal forms are canonical ways to represent any propositional logic formula using only AND, OR, and NOT.

They come in two dual flavors:

1. **Principal Disjunctive Normal Form (PDNF):**

   A sum-of-products built from all true assignments.

2. **Principal Conjunctive Normal Form (PCNF):**

   A product-of-sums built from all false assignments.

15.1 Minterms and Principal Disjunctive Normal Form (PDNF)

Intuitive Idea

"Lock & Key"

Imagine a safe with a combination lock that only opens when you hold exactly the right set of keys.

Here, each key is like a variable (say p or q), and having or not having a key corresponds to True or False.

"Minterm"

A minterm is a very specific "recipe" of keys — all keys listed, either you have them (True) or you do not (False) — that makes the lock open.

"Principal Disjunctive Normal Form (PDNF)"

When we OR together every recipe that works, we get the Principal Disjunctive Normal Form (PDNF).

15.2 Maxterms and Principal Conjunctive Normal Form (PCNF)

Intuitive Idea

"Lock & Key"

Imagine a safe with a combination lock that only opens when you hold exactly the right set of keys.

Here, each key is like a variable (say p or q), and having or not having a key corresponds to True or False.

"Maxterm"

A maxterm is a very specific "barrier" of keys — that blocks the lock from opening.

Each barrier lists all variables, joined by OR, describing a situation where the formula is false.

"Principal Conjunctive Normal Form (PCNF)"

When we AND together every barrier (maxterm), we get the Principal Conjunctive Normal Form (PCNF).

15.3 Minterms vs Maxterms

Aspect	**Minterm**	**Maxterm**
Built from	True rows	False rows
Clause form	AND of literals	OR of literals
Combined via	OR	AND
Purpose	Lists satisfying cases	Blocks falsifying cases
Used in	PDNF	PCNF

Example 22. Without constructing the truth tables, find the principal disjunctive normal forms of the following statements:

1. $$(\neg p \to q) \land (q \leftrightarrow p)$$
2. $$(p \land q) \lor (\neg p \land q) \lor (q \land r)$$
3. $$p \land \neg(q \land r) \lor (p \to q)$$
4. $$(q \lor (p \land r)) \land \neg((p \lor r) \land q)$$

Solution (1)

$$(\neg p \to q) \land (q \leftrightarrow p)$$

$$\equiv (p \lor q) \land ((p \land q) \lor (\neg p \land \neg q))$$

$$\equiv ((p \lor q) \land (p \land q)) \lor ((p \lor q) \land \neg(p \lor q))$$

$$\equiv ((p \lor q) \land (p \land q)) \lor F$$

$$\equiv (p \wedge (p \wedge q)) \vee (q \wedge (p \wedge q))$$

$$\equiv (p \wedge q) \vee (p \wedge q)$$

$$\equiv p \wedge q$$

Solution (2)

$$(p \wedge q) \vee (\neg p \wedge q) \vee (q \wedge r)$$

$$\equiv ((p \wedge q) \wedge (r \vee \neg r)) \vee ((\neg p \wedge q) \wedge (r \vee \neg r)) \vee ((q \wedge r) \wedge (p \vee \neg p))$$

$$\equiv (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r)$$

$$\vee (p \wedge q \wedge r) \vee (\neg p \wedge q \wedge r)$$

$$\equiv (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r)$$

(Deleting repetition of identical minterms)

Solution (3)

$$p \wedge \neg(q \wedge r) \vee (p \rightarrow q)$$

$$\equiv (p \wedge (\neg q \vee \neg r)) \vee (\neg p \vee q)$$

$$\equiv (p \wedge \neg q) \vee (p \wedge \neg r) \vee \neg p \vee q$$

$$\equiv (p \wedge \neg q) \vee (p \wedge \neg r) \vee (\neg p \wedge (q \vee \neg q)) \vee (q \wedge (p \vee \neg p))$$

$$\equiv (p \wedge \neg q) \vee (p \wedge \neg r) \vee (\neg p \wedge q) \vee (\neg p \wedge \neg q) \vee (p \wedge q) \vee (\neg p \wedge q)$$

$$\equiv ((p \wedge \neg q) \wedge (r \vee \neg r)) \vee ((p \wedge \neg r) \wedge (q \vee \neg q))$$

$$\vee ((\neg p \wedge q) \wedge (r \vee \neg r)) \vee ((p \wedge q) \wedge (r \vee \neg r))$$

$$\equiv (p \wedge \neg q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \vee (p \wedge q \wedge \neg r)$$

$$\vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \vee (p \wedge q \wedge r)$$

$$\vee(p \wedge q \wedge \neg r)$$

$$\equiv (p \wedge \neg q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \vee (p \wedge q \wedge \neg r)$$

$$\vee(\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \vee (p \wedge q \wedge r)$$

(Omitting repetitions)

Note 5. Since all possible minterms are present in the PDNF, we infer that the given statement is a tautology.

Solution (4)

$$(q \vee (p \wedge r)) \wedge \neg((p \vee r) \wedge q)$$

$$\equiv (q \vee (p \wedge r)) \wedge (\neg(p \vee r) \vee \neg q)$$

$$\equiv (q \vee (p \wedge r)) \wedge ((\neg p \wedge \neg r) \vee \neg q)$$

$$\equiv (q \wedge \neg p \wedge \neg r) \vee (q \wedge \neg q) \vee (p \wedge r \wedge \neg p \wedge \neg r) \vee (p \wedge r \wedge \neg q)$$

$$\equiv (\neg p \wedge q \wedge \neg r) \vee F \vee F \vee (p \wedge \neg q \wedge r)$$

$$\equiv (\neg p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r)$$

(Deleting F's)

16 Try yourself

1. Construct the truth table for each of the following compound propositions:
   (a) $(p \wedge q) \rightarrow (p \vee q)$
   (b) $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$
   (c) $(p \rightarrow q) \vee (\neg p \rightarrow q)$
   (d) $(p \rightarrow q) \wedge (\neg p \rightarrow q)$
   (e) $(p \leftrightarrow q) \vee (\neg p \leftrightarrow q)$
2. Determine which of the following statements are tautologies or contradictions.
   (a) $(p \rightarrow \neg p) \rightarrow \neg p$
   (b) $p \rightarrow (p \vee q)$
   (c) $(\neg q \rightarrow p) \wedge q$

(d) $(q \to p) \wedge (\neg p \wedge q)$

3. Prove the following equivalences:

   (a) $\neg(p \to q) \equiv p \wedge \neg q$

   (b) $(p \wedge q) \vee (p \wedge \neg q) \equiv p$

   (c) $(p \vee q) \wedge (p \vee \neg q) \equiv p$

   (d) $\neg(p \leftrightarrow q) \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$

4. Write down the duals of the following statements:

   (a) $\neg(p \vee q) \vee [(\neg p) \wedge q] \vee p$

   (b) $\neg p \to (p \to q)$

   (c) $(p \wedge q) \to (p \to q)$

   (d) $(p \to q) \to (\neg q \to \neg p)$

5. Prove the following implications, using truth tables:

   (a) $(p \wedge q) \Rightarrow (p \to q)$

   (b) $(q \to p) \Rightarrow (\neg p \to \neg q)$

   (c) $p \to q \Rightarrow p \to (p \wedge q)$

   (d) $(p \to q) \to q \Rightarrow p \vee q$

6. Find a DNF or a CNF of the following:

   (a) $(p \wedge (p \to q))$

   (b) $\neg(p \to q)$

   (c) $\neg(p \leftrightarrow q)$

   (d) $q \wedge (p \vee \neg q)$

7. Find the PDNF of the following statements using truth tables:

   (a) $p \wedge (p \to q)$

   (b) $\neg(p \vee q) \leftrightarrow (p \wedge q)$

   (c) $q \wedge (p \vee \neg q)$

   (d) $(q \to p) \wedge (\neg p \wedge q)$

8. Construct the truth table for each of the following compound propositions:

   (a) $(\neg p \wedge (\neg q \wedge r)) \vee (q \wedge r) \vee (p \wedge r)$

   (b) $((p \to q) \wedge (q \to r)) \to (p \to r)$

   (c) $((p \vee q) \wedge ((p \to r) \wedge (q \to r))) \to r$

   (d) $(p \leftrightarrow q) \vee (\neg q \leftrightarrow r)$

   (e) $(p \leftrightarrow q) \leftrightarrow (r \leftrightarrow s)$

9. Determine which of the following statements are tautologies or contradictions:

(a) $(p \wedge q) \wedge \neg(p \vee q)$

(b) $q \vee (p \wedge \neg q) \vee (\neg p \wedge \neg q)$

(c) $((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$

(d) $((p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r)))$

(e) $((p \wedge q) \vee \neg(\neg p \vee (\neg q \wedge \neg r))) \wedge (\neg p \vee \neg q) \wedge (\neg p \vee \neg r)$

10. By constructing truth tables, prove the following equivalences:

(a) $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$

(b) $q \wedge (p \vee \neg q) \wedge (\neg p \vee \neg q) \equiv F$

(c) $(p \rightarrow q) \wedge (r \rightarrow q) \equiv (p \vee r) \rightarrow q$

11. Without using truth tables, prove the following equivalences:

(a) $(p \vee q) \wedge (\neg p \wedge (\neg p \wedge q)) \equiv (\neg p \wedge q)$

(b) $(\neg p \wedge (\neg q \wedge r)) \vee (q \wedge r) \vee (p \wedge r) \equiv r$

(c) $p \rightarrow (q \vee r) \equiv (p \rightarrow q) \vee (p \rightarrow r)$

(d) $(p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$

(e) $(\neg p \rightarrow \neg q) \rightarrow (q \rightarrow p) \equiv T$

12. Write down the duals of the following equivalences and prove the duals without using truth tables:

(a) $\neg(p \wedge q) \rightarrow (\neg p \vee (\neg p \vee q)) \equiv (\neg p \vee q)$

(b) $(\neg p \rightarrow (\neg p \rightarrow (\neg p \wedge q))) \equiv p \vee q$

(c) $p \leftrightarrow q \equiv (p \vee q) \rightarrow (p \wedge q)$

(d) $(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$

(e) $\neg p \rightarrow (q \rightarrow r) \equiv q \rightarrow (p \vee r)$

13. Prove the following implications, using truth tables:

(a) $((p \vee \neg(q \wedge r)) \wedge \neg p) \Rightarrow (\neg q \vee \neg r)$

(b) $(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow s) \Rightarrow (s \vee r)$

14. Prove the following implications, without using truth tables:

(a) $(p \rightarrow q) \wedge (q \rightarrow r) \Rightarrow (p \rightarrow r)$

(b) $(q \rightarrow (p \wedge \neg p)) \rightarrow (r \rightarrow (p \wedge \neg p)) \Rightarrow (r \rightarrow q)$

15. Find the DNF of the following statements:

(a) $\neg(p \rightarrow (q \wedge r))$

(b) $(\neg p \rightarrow r) \wedge (p \leftrightarrow q)$

(c) $(q \vee (p \wedge r)) \wedge \neg((p \vee r) \wedge q)$

16. Find the CNF of the following statements:

(a) $\neg(p \vee q) \leftrightarrow (p \wedge q)$

(b) $\neg((p \vee \neg q) \wedge \neg r)$

(c) $q \vee (p \wedge r) \wedge \neg((p \vee r) \wedge q)$

17. Find the PDNF and PCNF of the following statements using truth tables:

(a) $p \rightarrow (p \wedge (q \rightarrow p))$

(b) $(q \rightarrow p) \wedge (\neg p \wedge q)$

(c) $(p \wedge q) \vee (p \wedge r) \vee (q \wedge r)$

(d) $(p \wedge q) \vee (\neg p \wedge q) \vee (q \wedge r)$

18. Without using truth tables, find the PDNF of the following statements:

(a) $(p \wedge q) \vee (\neg p \wedge r) \vee (q \wedge r)$

(b) $p \rightarrow ((p \rightarrow q) \wedge \neg(\neg q \vee \neg p))$

(c) $(\neg((p \vee q) \wedge r)) \wedge (p \vee r)$

(d) $(p \wedge \neg q) \vee (q \wedge \neg p) \vee (r \wedge p)$

19. Without using truth tables, find the PCNF of the following statements:

(a) $(\neg p \rightarrow r) \wedge (q \leftrightarrow p)$

(b) $p \wedge (p \rightarrow q) \leftrightarrow p \wedge (\neg p \vee q)$

(c) $p \vee (\neg p \wedge \neg q \wedge r)$

(d) $p \vee (\neg p \rightarrow (q \vee (\neg q \rightarrow r)))$

20. If $A(x)$: x is an animal, $B(x)$: x is black and $C(x)$: x is a cat, translate the following in words:

(a) $\forall x[C(x) \rightarrow A(x)]$

(b) $\exists x[C(x) \wedge B(x)]$

21. Show that $\forall x\ P(x) \rightarrow \exists x\ P(x)$ is a logically valid statement.

22. Show that

$$\forall x\ (P(x) \wedge Q(x)) \leftrightarrow \forall x\ P(x) \wedge \forall x\ Q(x)$$

is a logically valid statement.

23. Show that

$$\exists x\ (P(x) \vee Q(x)) \leftrightarrow \exists x\ P(x) \vee \exists x\ Q(x)$$

is a logically valid statement.

24. Show that

$$\forall x\ P(x) \vee \forall x\ Q(x) \rightarrow \forall x\ (P(x) \vee Q(x))$$

is a valid statement. Is the statement

$$\forall x\ (P(x) \vee Q(x)) \rightarrow \forall x\ P(x) \vee \forall x\ Q(x)$$

valid?

25. Show that the premises "Everyone in the Computer Science branch has studied Discrete Mathematics" and "Ram is in Computer Science branch" imply that "Ram has studied Discrete Mathematics".

26. Show that
$$\neg[\exists x\ P(x) \wedge Q(a)] \Rightarrow \exists x\ P(x) \rightarrow \neg Q(a)$$

27. Show that $\neg P(a, b)$ follows logically from
$$\forall x\ \forall y\ (P(x, y) \rightarrow Q(x, y))$$
and
$$\neg Q(a, b).$$

28. Show that the conclusion C follows from the premises H_1, H_2, H_3 in the following cases using truth table technique:
    (a) $H_1 : p \rightarrow (q \rightarrow r), \quad H_2 : p \wedge q, \quad C : r$
    (b) $H_1 : \neg p \rightarrow q, \quad H_2 : \neg(q \wedge \neg r), \quad H_3 : \neg r, \quad C : \neg p$

29. Prove the following by using direct method:
    (a) $p \vee q,\ p \rightarrow r,\ q \rightarrow s \Rightarrow s \vee r$
    (b) $a \wedge b,\ (a \leftrightarrow b) \rightarrow (c \vee d) \Rightarrow d \vee c$
    (c) $(p \wedge q) \rightarrow r,\ \neg r \vee s,\ \neg s \Rightarrow \neg p \vee \neg q$
    (d) $p \vee q,\ q \rightarrow r,\ p \rightarrow s,\ \neg s \Rightarrow r \wedge (p \vee q)$
    (e) $\neg(p \wedge \neg q),\ \neg q \vee r,\ \neg r \Rightarrow \neg p$
    (f) $p \rightarrow q,\ (\neg q \vee r) \wedge \neg r,\ \neg(\neg p \wedge s) \Rightarrow \neg s$
    (g) $(p \rightarrow q) \rightarrow r,\ p \wedge s,\ q \wedge t \Rightarrow r$
    (h) $\neg j \rightarrow (m \vee n),\ (h \vee g) \rightarrow \neg j,\ h \vee g \Rightarrow m \vee n$

30. Prove the following by using indirect method:
    (a) $p \rightarrow q,\ q \rightarrow r,\ \neg(p \wedge r),\ p \vee r \Rightarrow r$
    (b) $\neg q,\ p \rightarrow q,\ p \vee r \Rightarrow r$
    (c) $s \rightarrow \neg q,\ s \vee r,\ \neg r,\ \neg r \leftrightarrow q \Rightarrow \neg p$
    (d) $\neg(p \rightarrow q) \rightarrow \neg(r \vee s),\ (q \rightarrow p) \vee \neg r,\ r \Rightarrow p \leftrightarrow q$

31. Prove that each of the following sets of premises is inconsistent:
    (a) $p \rightarrow q,\ p \rightarrow r,\ q \rightarrow \neg r,\ p$
    (b) $p \rightarrow q,\ (q \vee r) \rightarrow s,\ s \rightarrow \neg p,\ p \wedge \neg r$
    (c) $p \rightarrow (q \rightarrow r),\ q \rightarrow (r \rightarrow s),\ p \wedge q \wedge \neg s$

32. Show that the following premises are inconsistent.
    (a) If Raja misses many classes, then he fails in the final examination.

(b) If Raja fails in the final examination, then he is uneducated.

(c) If Raja reads a lot of books, then he is not uneducated.

(d) Raja misses many classes and reads a lot of books.

33. "It is not sunny this afternoon and it is colder than yesterday"; "We will go to the playground only if it is sunny"; "If we do not go to the ground, then we will go to a movie" and "If we go to a movie, then we will return home by sunset" lead to the conclusion "We will return home by sunset".

34. Construct an argument using rules of inference to show that the hypotheses "Radha works hard", "If Radha works hard, then she is a dull girl" and "If Radha is a dull girl, then she will not get the job" imply the conclusion "Radha will not get the job".

35. "If I eat spicy food, then I have strange dreams". "I have strange dreams, if there is thunder while I sleep". "I did not have strange dreams". What relevant conclusion can be drawn from the above premises? Construct an argument to obtain your conclusion.

36. Show that the following set of premises is inconsistent.

    (a) John will get his degree, if and only if he passes all the examinations.

    (b) He will pass all the examinations, if and only if he works hard.

    (c) He will be unemployed, if and only if he does not get his degree.

    (d) John works hard if and only if he is employed.

37. If A works hard, then B or C will enjoy themselves. If B enjoys himself, then A will not work hard. If D enjoys himself, then C will not. Therefore, if A works hard, D will not enjoy himself.

    Translate the above into statements and prove the conclusion by using the CP-rule.

38. Symbolise the following expressions:

    (a) x is the father of the mother of y.

    (b) Everybody loves a lover.

39. Prove the following implications:

    (a) $\forall x\ (P(x) \to Q(x)) \wedge \forall x\ (Q(x) \to R(x)) \Rightarrow \forall x\ (P(x) \to R(x))$

    (b) $\exists x\ P(x),\ \forall x\ (P(x) \to Q(x)) \Rightarrow \exists x\ Q(x)$

    (c) $\exists x\ (P(x) \wedge Q(x)) \Rightarrow \exists x\ P(x) \wedge \exists x\ Q(x)$

    (d) $\exists x\ P(x) \to \forall x\ Q(x) \Rightarrow \forall x\ (P(x) \to Q(x))$

    (e) $\forall x\ (P(x) \to Q(x)) \Rightarrow \forall x\ P(x) \to \forall x\ Q(x)$

    (f) $\forall x\ (C(x) \to A(x)) \Rightarrow \forall x\Big((\exists y\ (C(y) \wedge B(x,y))) \to \exists y\ (A(y) \wedge B(x,y))\Big)$

17 Conclusion

Predicate calculus provides a rigorous mathematical framework for reasoning, formal verification and symbolic representation of knowledge. It forms the foundation of advanced topics in mathematics, computer science, artificial intelligence and formal systems.

Solved Examples from Past Examinations

1. (a) Check whether the compound proposition is tautology, contradiction, or contingency.

$$(p \vee q) \to (q \wedge r) \leftrightarrow (p \wedge \neg r)$$

Solution: To determine if the compound proposition $(p \vee q) \to (q \wedge r) \leftrightarrow (p \wedge \neg r)$ is a tautology, contradiction, or contingency, we need to construct a truth table. Since there are three variables (p, q, and r), the table will have $2^3 = 8$ rows.

p	q	r	$p \vee q$	$q \wedge r$	$(p \vee q) \to (q \wedge r)$	$\neg r$	$p \wedge \neg r$	$\leftrightarrow$
T	T	T	T	T	T	F	F	F
T	T	F	T	F	F	T	T	F
T	F	T	T	F	F	F	F	T
T	F	F	T	F	F	T	T	F
F	T	T	T	T	T	F	F	F
F	T	F	T	F	F	T	F	T
F	F	T	F	F	T	F	F	F
F	F	F	F	F	T	T	F	F

The final column contains both T and F values. Therefore, the proposition is a contingency.

(b) Show that

$$(p \wedge q) \wedge (p \to r) \wedge (q \to r) \to r$$

is a tautology without using truth tables.

Solution: To show that the proposition $(p \wedge q) \wedge (p \to r) \wedge (q \to r) \to r$ is a tautology, we can use the laws of propositional logic to simplify the expression until it reduces to T (True).

$$
\begin{aligned}
&((p \wedge q) \wedge (p \to r) \wedge (q \to r)) \to r \\
&\equiv ((p \wedge q) \wedge (\neg p \vee r) \wedge (\neg q \vee r)) \to r && \text{(Implication law } a \to b \equiv \neg a \vee b\text{)} \\
&\equiv \neg\big((p \wedge q) \wedge (\neg p \vee r) \wedge (\neg q \vee r)\big) \vee r && \text{(Implication law)} \\
&\equiv \neg(p \wedge q) \vee \neg(\neg p \vee r) \vee \neg(\neg q \vee r) \vee r && \text{(De Morgan's law)} \\
&\equiv (\neg p \vee \neg q) \vee (p \wedge \neg r) \vee (q \wedge \neg r) \vee r && \text{(De Morgan's law \& Involution law } \neg(\neg a) \equiv a\text{)} \\
&\equiv (\neg p \vee \neg q) \vee ((p \vee q) \wedge \neg r) \vee r && \text{(Distributive law)} \\
&\equiv (\neg p \vee \neg q) \vee ((p \vee q \vee r) \wedge (\neg r \vee r)) && \text{(Distributive law)} \\
&\equiv (\neg p \vee \neg q) \vee ((p \vee q \vee r) \wedge T) && \text{(Tautology law } \neg a \vee a \equiv T\text{)} \\
&\equiv (\neg p \vee \neg q) \vee (p \vee q \vee r) && \text{(Identity law)} \\
&\equiv (\neg p \vee p) \vee (\neg q \vee q) \vee r && \text{(Associative \& Commutative laws)} \\
&\equiv T \vee T \vee r && \text{(Tautology law)} \\
&\equiv T && \text{(Dominance law } a \vee T \equiv T\text{)}
\end{aligned}
$$

Hence, $((p \wedge q) \wedge (p \to r) \wedge (q \to r)) \to r$ is a tautology.

2. (a) Given the following two conditional statements:

   1. "If a student studies hard, then they will pass the exam."
   2. "If a shape is a square, then it has four equal sides."

   For each statement, write the:

   (i) Converse
   (ii) Inverse
   (iii) Contrapositive

   Solution: 1. "If a student studies hard, then they will pass the exam."

   Let, p: student studies hard, q: student passes exam. Then the given conditional statement is written as $p \to q$. Therefore,

   (i) Converse $q \to p$: If a student passes the exam, then they studied hard.
   (ii) Inverse $\neg p \to \neg q$: If a student does not study hard, then they will not pass the exam.
   (iii) Contrapositive $\neg q \to \neg p$: If a student does not pass the exam, then they did not study hard.

   2. "If a shape is a square, then it has four equal sides."

   Let, r: shape is square, s: shape has four equal sides. Then the given conditional statement is written as $r \to s$. Therefore,

   (i) Converse $s \to r$: If a shape has four equal sides, then it is a square.
   (ii) Inverse $\neg r \to \neg s$: If a shape is not a square, then it does not have four equal sides.
   (iii) Contrapositive $\neg s \to \neg r$: If a shape does not have four equal sides, then it is not a square.

(b) Use De Morgan's law to find the negative of the following statement.
'If it does not rain or if there is no traffic dislocation, then the sports day will be held and the cultural program will go on.'
Solution: Let,

p : It rains

q : There is traffic dislocation

r : Sports day is held

s : Cultural program goes on

Given statement: $(\neg p \vee \neg q) \rightarrow (r \wedge s)$

Negation: $\neg[(\neg p \vee \neg q) \rightarrow (r \wedge s)]$

Using Implication law $A \rightarrow B \equiv \neg A \vee B$:

$$\neg[(\neg p \vee \neg q) \rightarrow (r \wedge s)] \equiv \neg[\neg(\neg p \vee \neg q) \vee (r \wedge s)]$$

Applying De Morgan's law:

$$\neg[\neg(\neg p \vee \neg q) \vee (r \wedge s)] \equiv (\neg p \vee \neg q) \wedge (\neg r \vee \neg s)$$

$\therefore$ Negation $(\neg p \vee \neg q) \wedge (\neg r \vee \neg s)$: Either it does not rain or there is traffic dislocation, and either the sports day is not held or the cultural program does not go on.

3. (a) Prove that

$$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r.$$

(i) With truth table

(ii) Without truth table

Solution:

(i) *Proof with truth table*:

p	q	r	$p \rightarrow r$	$q \rightarrow r$	LHS: $(p \rightarrow r) \vee (q \rightarrow r)$	$p \wedge q$	RHS: $(p \wedge q) \rightarrow r$
T	T	T	T	T	T	T	T
T	T	F	F	F	F	T	F
T	F	T	T	T	T	F	T
T	F	F	F	T	T	F	T
F	T	T	T	T	T	F	T
F	T	F	T	F	T	F	T
F	F	T	T	T	T	F	T
F	F	F	T	T	T	F	T

Since the truth values of LHS and RHS are same in all rows, the propositions are logically equivalent, i.e.,

$$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

(ii) *Proof without truth table*:
To prove the equivalence, we apply the laws of propositional logic to the Left-Hand Side (LHS):

$$\begin{aligned}
\text{LHS} &= (p \to r) \vee (q \to r) \\
&\equiv (\neg p \vee r) \vee (\neg q \vee r) && \text{(Implication law)} \\
&\equiv (\neg p \vee \neg q) \vee (r \vee r) && \text{(Associative \& Commutative laws)} \\
&\equiv (\neg p \vee \neg q) \vee r && \text{(Idempotent law)} \\
&\equiv \neg(p \wedge q) \vee r && \text{(De Morgan's law)} \\
&\equiv (p \wedge q) \to r && \text{(Implication law)} \\
&= \text{RHS}
\end{aligned}$$

(b) Express the following propositions X, Y symbolically, and hence determine whether X and Y are logically equivalent or not by using truth table.

X: If Alice is both smart and honest, then she is not intelligent.

Y: If Alice either smart or not intelligent, then she is not honest.

Solution: Let,

p : Alice is smart

q : Alice is honest

r : Alice is intelligent

Symbolic Representation:

Proposition X: "If Alice is both smart and honest, then she is not intelligent."

$$X : (p \wedge q) \to \neg r$$

Proposition Y: "If Alice either smart or not intelligent, then she is not honest."

$$Y : (p \vee \neg r) \to \neg q$$

Truth table comparison for propositions X and Y:

p	q	r	$p \wedge q$	$\neg r$	$X : (p \wedge q) \to \neg r$	$p \vee \neg r$	$\neg q$	$Y : (p \vee \neg r) \to \neg q$
T	T	T	T	F	F	T	F	F
T	T	F	T	T	T	T	F	F
T	F	T	F	F	T	T	T	T
T	F	F	F	T	T	T	T	T
F	T	T	F	F	T	F	F	T
F	T	F	F	T	T	T	F	F
F	F	T	F	F	T	F	T	T
F	F	F	F	T	T	T	T	T

Since the truth values of X and Y are not the same in all rows, the propositions are not logically equivalent, i.e.,

$$(p \wedge q) \to \neg r \not\equiv (p \vee \neg r) \to \neg q$$

4. (a) Discuss the consistency of the following statements:

- If RCB lose many matches, then they will not reach the final.
- If RCB do not reach the final, then they will not be champions.
- If RCB reach the final, then they will be champions.
- If Kohli scores a lot of runs, then RCB will be champions.
- RCB lose many matches and Kohli scores a lot of runs.

Solution: Let,

L : RCB lose many matches

F : RCB reach the final

C : RCB are champions

K : Kohli scores a lot of runs

Then the given statements are written as:

1. $L \to \neg F$
2. $\neg F \to \neg C$
3. $F \to C$
4. $K \to C$
5. $L \wedge K$

Now if we derive a contradiction by using the given statements, then they are inconsistent.

6. L	Using 5 and Simplification
7. $\neg F$	Modus Ponens using 1 and 6
8. $\neg C$	Modus Ponens using 2 and 7
9. K	Using 5 and Simplification
10. C	Modus Ponens using 4 and 9
11. $C \wedge \neg C$	Conjunction using 8 and 10
12. F	Using 11 and Contradiction

$\therefore$ The statements are inconsistent.

(b) Prove that the premises

$$p \to q, \quad q \to r, \quad s \to \sim r, \quad \text{and} \quad q \wedge s$$

are inconsistent.

Solution: To prove that the given premises are inconsistent, we derive a

contradiction.

1. $q \wedge s$	Premise (Given)
2. q	Using 1 and Simplification
3. s	Using 1 and Simplification
4. $q \rightarrow r$	Premise (Given)
5. r	Modus Ponens using 2 and 4
6. $s \rightarrow \sim r$	Premise (Given)
7. $\sim r$	Modus Ponens using 3 and 6
8. $r \wedge \sim r$	Conjunction using 5 and 7
9. F	8 and Contradiction

$\therefore$ The premises are inconsistent.

5. (a) Show that t is a valid conclusion from the premises:

$$p \rightarrow q, \quad q \rightarrow r, \quad r \rightarrow s, \quad \sim s, \quad \text{and} \quad p \vee t.$$

Solution: The valid argument for deducing t from the given premises is given as a sequence.

1. $p \longrightarrow q$	Premise (Given)
2. $q \longrightarrow r$	Premise (Given)
3. $r \longrightarrow s$	Premise (Given)
4. $\sim s$	Premise (Given)
5. $p \vee t$	Premise (Given)
6. $p \longrightarrow r$	Hypothetical Syllogism using 1 and 2
7. $p \longrightarrow s$	Hypothetical Syllogism using 6 and 3
8. $\sim p$	Modus Tollens using 7 and 4
9. t	Disjunctive Syllogism using 5 and 8

$\therefore$ The conclusion t is valid.

(b) Prove that the following arguments are valid without using truth tables:

(i) $p \rightarrow \sim q,\ r \rightarrow q,\ r \ \therefore \sim p$

(ii) $p \vee (q \rightarrow p),\ \sim p \wedge r \ \therefore \sim q$

Solution:

(i) The valid argument for deducing $\sim p$ from the given premises is given as a sequence.

1. $p \longrightarrow \sim q$	Premise (Given)
2. $r \longrightarrow q$	Premise (Given)
3. r	Premise (Given)
4. q	Modus Ponens using 2 and 3
5. $\sim\sim q$	Using 4 and Involution law
6. $\sim p$	Modus Tollens using 1 and 5

$\therefore$ The conclusion $\sim p$ is valid.

(ii) The valid argument for deducing $\sim q$ from the given premises is given as a sequence.

1. $p \vee (q \longrightarrow p)$	Premise (Given)
2. $\sim p \wedge r$	Premise (Given)
3. $\sim p$	Using 2 and Simplification
4. $q \longrightarrow p$	Disjunctive Syllogism using 1 and 3
5. $\sim q$	Modus Tollens using 4 and 3

$\therefore$ The conclusion $\sim q$ is valid.

6. (a) Examine the validity of the following argument: "If the market is free then there is no inflation. If there is no inflation then there are price controls. Since there are price controls, therefore market is free."
Solution: Let,

p : The market is free

q : There is inflation

r : There are price controls

Symbolic Representation:
Given premises: $p \longrightarrow \sim q$, $\sim q \longrightarrow r$, r
Conclusion: p
From the given premises, the following can be deduced.

1. $p \longrightarrow \sim q$	Premise (Given)
2. $\sim q \longrightarrow r$	Premise (Given)
3. r	Premise (Given)
4. $p \longrightarrow r$	Hypothetical Syllogism using 1 and 2

But from r and $p \longrightarrow r$, we cannot conclude p. This is the fallacy of Affirming the Conclusion.
Therefore, the argument is invalid.

p	r	Premise 1: $p \longrightarrow r$	Premise 2: r	Conclusion: p
T	T	T	T	T
T	F	F	F	T
F	***T***	***T***	***T***	***F* (FAIL)**
F	F	T	F	F

Table 1: Truth Table for Affirming the Conclusion

(b) You are about to leave for school in the morning and discover that you do not have your glasses. You know the following statements are true:

a. If I was reading the newspaper in the kitchen, then my glasses are on the kitchen table.

b. If my glasses are on the kitchen table, then I saw them at breakfast.

c. I did not see my glasses at breakfast.

d. I was reading the newspaper in the living room or I was reading the newspaper in the kitchen.

e. If I was reading the newspaper in the living room, then my glasses are on the coffee table.

Use the rules of inference to find where the glasses are.
Solution: Let,

K : I was reading the newspaper in the kitchen

L : I was reading the newspaper in the living room

T : My glasses are on the kitchen table

B : I saw my glasses at breakfast

C : My glasses are on the coffee table

Symbolic Representation:
Given statements: $K \longrightarrow T$, $T \longrightarrow B$, $\sim B$, $L \vee K$, $L \longrightarrow C$
Now apply rules of inference.

1. $K \rightarrow T$	Premise (Given)
2. $T \rightarrow B$	Premise (Given)
3. $\sim B$	Premise (Given)
4. $L \vee K$	Premise (Given)
5. $L \rightarrow C$	Premise (Given)
6. $\sim T$	Modus Tollens using 2 and 3
7. $\sim K$	Modus Tollens using 1 and 6
8. L	Disjunctive Syllogism using 4 and 7
9. C	Modus Ponens using 5 and 8

Therefore, the glasses are on the coffee table.

7. Show the validity of the following argument using the rule of conditional proof.

"If my program runs successfully then I will submit my project. I can appear in the examination if I submit my project. Either my program runs successfully or the computer crashes. Therefore, if the computer does not crash then I can appear at the examination."
Solution: Let,

P : My program runs successfully

S : I submit my project

E : I can appear in the examination

C : The computer crashes

Symbolic Representation:
Given premises: $P \rightarrow S$, $S \rightarrow E$, $(P \wedge \neg C) \vee (\neg P \wedge C)$

Conclusion: $\neg C \to E$
In conditional proof, we assume $\neg C$ as an additional premise and try to derive conclusion E.
Now apply rules of inference.

1. $P \to S$	Premise (Given)
2. $S \to E$	Premise (Given)
3. $(P \wedge \neg C) \vee (\neg P \wedge C)$	Premise (Given)
4. $\neg C$	Additional premise for conditional proof
5. $P \to E$	Hypothetical Syllogism using 1 and 2
6. $\neg C \vee P$	Using 4 and Addition
7. $\neg(C \wedge \neg P)$	Using 6, Involution law and De Morgan's law
8. $P \wedge \neg C$	Disjunctive Syllogism using 3 and 7
9. P	Using 8 and Simplification
10. E	Modus Ponens using 5 and 9

Hence, the argument is valid.

8. Write the following statements using nested quantifiers:

   (i) Every street in the city has a house in which we can find a person who is either rich or highly educated.

   (ii) There is a room in Academic Building I where all the students of VIT Bhopal can sing and dance.

   (iii) Given any student in the campus, the student has a friend who has a job in Pune.

   Solution:

   (i) Every street in the city has a house in which we can find a person who is either rich or highly educated. Let,

   $$S(x) : x \text{ is a street in the city}$$
   $$H(y, x) : y \text{ is a house on street } x$$
   $$P(z, y) : z \text{ is a person in house } y$$
   $$R(z) : z \text{ is rich}$$
   $$E(z) : z \text{ is highly educated}$$

   Symbolic form: $\forall x \left(S(x) \to \exists y \left(H(y, x) \wedge \exists z \left(P(z, y) \wedge (R(z) \vee E(z))\right)\right)\right)$

   (ii) There is a room in Academic Building I where all the students of VIT Bhopal can sing and dance.
   Let,

   $$R(x) : x \text{ is a room in Academic Building I}$$
   $$S(y) : y \text{ is a student of VIT Bhopal}$$
   $$C(y, x) : \text{student } y \text{ can sing and dance in room } x$$

   Symbolic form: $\exists x \left(R(x) \wedge \forall y \left(S(y) \to C(y, x)\right)\right)$

(iii) Given any student in the campus, the student has a friend who has a job in Pune.
Let,

$$S(x) : x \text{ is a street in the campus}$$
$$F(x, y) : y \text{ is a friend of } x$$
$$J(y) : y \text{ has a job in Pune}$$

Symbolic form: $\forall x \left(S(x) \to \exists y \left(F(x, y) \wedge J(y)\right)\right)$

9. Translate each of these nested quantifications into an English statement that expresses a mathematical fact. The domain in each case consists of all real numbers.

(i) $\exists x \, \forall y \, (x + y = y)$

(ii) $\forall x \, \forall y \left(((x \geq 0) \wedge (y < 0)) \to (x - y > 0)\right)$

(iii) $\exists x \, \exists y \left(((x \leq 0) \wedge (y \leq 0)) \wedge (x - y > 0)\right)$

(iv) $\forall x \, \forall y \left(((x \neq 0) \wedge (y \neq 0)) \leftrightarrow (xy \neq 0)\right)$

Solution:

(i) $\exists x \, \forall y \, (x + y = y)$
Translation: There exists a real number x such that for every real number y, the sum of x and y equals y.
(Equivalently: There is a real number that is an additive identity in the set of real numbers.)

(ii) $\forall x \, \forall y \left(((x \geq 0) \wedge (y < 0)) \to (x - y > 0)\right)$
Translation: For every pair of real numbers x and y, if x is non-negative and y is negative, then the difference $x - y$ is positive.

(iii) $\exists x \, \exists y \left(((x \leq 0) \wedge (y \leq 0)) \wedge (x - y > 0)\right)$
Translation: There exist non-positive real numbers x and y such that their difference $x - y$ is positive.

(iv) $\forall x \, \forall y \left(((x \neq 0) \wedge (y \neq 0)) \leftrightarrow (xy \neq 0)\right)$
Translation: For all real numbers x and y, both x and y are non-zero if and only if their product xy is non-zero.

10. Determine the truth value and negation of the following statements, where the universal set is $U = \{1, 2, 3\}$:

(i) $\exists a \, \forall b, \; a^2 - 1 < b$

(ii) $\forall a \, \exists b, \; a^2 + b^2 < 17$

(iii) $\forall a \, \forall b, \; a^2 + b^2 < 15$

Solution:

(i) Statement: $\exists a \, \forall b, \; a^2 - 1 < b$
This statement asserts there exists at least one element $a \in U$ such that for every element $b \in U$, the inequality $a^2 - 1 < b$ holds.

- Let $a = 1$. Then $a^2 - 1 = 1^2 - 1 = 0$.
- For $b = 1$: $0 < 1$ (True)
- For $b = 2$: $0 < 2$ (True)
- For $b = 3$: $0 < 3$ (True)

Since the condition holds for $a = 1$, the statement is satisfied.

Truth Value: True
Negation: $\neg(\exists a\, \forall b,\, a^2 - 1 < b) \equiv \forall a\, \exists b,\, a^2 - 1 \geq b$

(ii) Statement: $\forall a\, \exists b,\, a^2 + b^2 < 17$
This statement asserts that for every $a \in U$, there is at least one $b \in U$ such that $a^2 + b^2 < 17$.

- If $a = 1$: Choose $b = 1 \implies 1^2 + 1^2 = 2 < 17$ (True)
- If $a = 2$: Choose $b = 1 \implies 2^2 + 1^2 = 5 < 17$ (True)
- If $a = 3$: Choose $b = 1 \implies 3^2 + 1^2 = 10 < 17$ (True)

The condition holds for every a.

Truth Value: True
Negation: $\neg(\forall a\, \exists b,\, a^2 + b^2 < 17) \equiv \exists a\, \forall b,\, a^2 + b^2 \geq 17$

(iii) Statement: $\forall a\, \forall b,\, a^2 + b^2 < 15$
This statement asserts that the sum of squares is less than 15 for every possible pair (a, b).

- Counterexample: Let $a = 3$ and $b = 3$.
- $a^2 + b^2 = 3^2 + 3^2 = 9 + 9 = 18$.
- $18 < 15$ is False.

Since at least one pair fails, the universal statement is false.

Truth Value: False
Negation: $\neg(\forall a\, \forall b,\, a^2 + b^2 < 15) \equiv \exists a\, \exists b,\, a^2 + b^2 \geq 15$

11. A doctor is analyzing a patient's symptoms. Given the following premises:

- If a person has a fever, then they have a disease.
- If a person has chills, then they have a fever.
- Alice has chills.
- If a person has a disease, then they need medication.
- If a person needs medication, then they are not healthy.

Using predicate logic, prove that Alice is not healthy.
Solution: Let,

$$C(x) : x \text{ has chills}$$
$$F(x) : x \text{ has fever}$$
$$D(x) : x \text{ has a disease}$$
$$M(x) : x \text{ needs medication}$$
$$H(x) : x \text{ is healthy}$$

Symbolic Representation:

Given premises : $\forall x\,(F(x) \to D(x))$, $\forall x\,(C(x) \to F(x))$, $C(\text{Alice})$, $\forall x\,(D(x) \to M(x))$, $\forall x\,(M(x) \to \neg H(x))$

Conclusion: $\neg H(\text{Alice})$
Now apply rules of inference.

1. $\forall x\,(F(x) \to D(x))$	Premise (Given)
2. $\forall x\,(C(x) \to F(x))$	Premise (Given)
3. $C(\text{Alice})$	Premise (Given)
4. $\forall x\,(D(x) \to M(x))$	Premise (Given)
5. $\forall x\,(M(x) \to \neg H(x))$	Premise (Given)
6. $C(\text{Alice}) \to F(\text{Alice})$	Universal Instantiation from 2
7. $F(\text{Alice})$	Modus Ponens using 3 and 6
8. $F(\text{Alice}) \to D(\text{Alice})$	Universal Instantiation from 1
9. $D(\text{Alice})$	Modus Ponens using 7 and 8
10. $D(\text{Alice}) \to M(\text{Alice})$	Universal Instantiation from 4
11. $M(\text{Alice})$	Modus Ponens using 9 and 10
12. $M(\text{Alice}) \to \neg H(\text{Alice})$	Universal Instantiation from 5
13. $\neg H(\text{Alice})$	Modus Ponens using 11 and 12

Therefore, Alice is not healthy is a valid conclusion.

12. Verify the validity of the inference.

    If one person is more successful than another, then he has worked harder to deserve success. John has not worked harder than Peter. Therefore, John is not successful than Peter.

    Solution: Let,

    $S(x, y) : x$ is more successful than y
    $W(x, y) : x$ has worked harder than y to deserve success
    $j :$ John
    $p :$ Peter

    Symbolic Representation:
    Given premises: $\forall x\,\forall y\,[S(x, y) \to W(x, y)]$, $\neg W(j, p)$
    Conclusion: $\neg S(j, p)$
    Now apply rules of inference.

1. $\neg W(j, p)$	Premise (Given)
2. $\forall x\,\forall y\,[S(x, y) \to W(x, y)]$	Premise (Given)
3. $\forall y\,[S(j, y) \to W(j, y)]$	Universal instantiation and 2
4. $S(j, p) \to W(j, p)$	Universal instantiation and 3
5. $\neg S(j, p)$	Modus Tollens using 1 and 4

    Hence, the argument is valid.

Exercise Problems from Past Examinations

1. Determine whether $(p \vee q) \wedge (\neg p \vee r) \rightarrow (q \vee r)$ is a tautology or contradiction using laws of statement calculus. Mention all the laws properly in each steps.

   [Ans. Tautology]

2. Test the validity of the following argument. If I will select in IAS examination, then I will not be able to go to London. Since, I am going to London, I will not select in IAS examination. [Ans. Valid]

3. Prove the validity of the following argument "If I get the job and work hard, then I will get promoted. If I get promoted, then I will be happy. I will not be happy. Therefore, I will not get the job or I will not work hard". [Ans. Valid]

4. Show that $\sim p$ is a valid conclusion from the premises $(p \rightarrow q),\ (r \rightarrow s),\ (q \rightarrow t),\ (s \rightarrow u),\ \sim (t \wedge u)$, and $(p \rightarrow r)$.

5. State the converse, inverse, and contrapositive of the following statement. 'A positive integer is a prime only if it has no divisors other than 1 and itself.'

6. Write the contrapositive of the statement "(a, b) is related to (c, d) if a is related to c or if a is equal to c, then b is related to d".

7. (a) "If a number is divisible by 10 , then the number ends in zero". Write the given statement into symbolic form and then write contrapositive of the statement.

   (b) Prove or Disprove: $(p \wedge q) \wedge (\sim (p \vee q))$ is a tautology. [Ans. Contradiction]

8. Verify the validity of the following argument using inference rules: "If I take the day off, it either rains or snows." "I took Tuesday off or I took Thursday off." "It was sunny on Tuesday." "It did not snow on Thursday." Therefore, "It rained on Thursday." [Ans. Valid]

9. (a) Using truth table find whether the following expression is a tautology, contradiction, or contingency:

   $$(p \vee q) \rightarrow (p \wedge \neg q) \leftrightarrow (\neg p \vee q)$$

   [Ans. Contingency]

   (b) Prove the following equivalence without using truth table:

   $$(p \rightarrow q) \rightarrow (q \rightarrow r) \equiv \neg q \vee r$$

10. Show that $(t \wedge s)$ can be derived from the premises $p \rightarrow q,\ q \rightarrow \sim r,\ r,\ p \vee (t \wedge s)$.

11. (a) Check whether the given compound proposition

    $$\neg(q \rightarrow r) \wedge r \wedge (p \rightarrow q)$$

    is a tautology, contingency or contradiction. Justify your answer.

    [Ans. Contradiction]

(b) State the converse, contrapositive and inverse of the following conditional statement. "I will go to the beach whenever it is a sunny summer day."

12. (a) A university database has a table Students (Name, Course, Grade), where Name represents the student's name, Course represents the course the student is enrolled in, and Grade represents the student's grade in that course. The university has the following policies:

(A) Every student must pass at least one course.

(B) If a student passes a course, they are eligible for the next level of that course.

(C) A student is placed on academic probation if they pass fewer than two courses.

Answer the following:

(i) Express all policies in symbolic form.

(ii) Negate all policy statements and provide their symbolic notation.

(b) Given the following premises:

(i) If a student studies consistently, they will understand the material.

(ii) If a student understands the material, they will pass the exam.

(iii) If a student does not pass the exam, they must retake the course.

(iv) Sofia studied consistently.

Can we logically conclude that Sofia does not have to retake the course? Is the conclusion valid? Justify your answer. [Ans. Invalid]

13. (a) Check whether $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are logically equivalent? [Ans. Equivalent]

(b) Prove without using truth table: $\neg(p \leftrightarrow q) \equiv \neg p \leftrightarrow q$.

14. Show that the premises "It is not sunny this afternoon and it is colder than yesterday," "We will go swimming only if it is sunny," "If we do not go swimming, then we will take a canoe trip," and "If we take a canoe trip, then we will be home by sunset" lead to the conclusion "We will be home by sunset."

15. (a) Show that $\neg r$ is a valid conclusion from the premises $p \to \neg q, \quad r \to p, \quad q$

(i) with truth table

(ii) without truth table.

(b) Write the English sentence using the logical expression.
"You cannot ride the roller coaster if you are under 4 feet tall unless you are older than 16 years old."

16. (a) Without using the truth table show that

$$\sim (p \vee q) \leftrightarrow (p \wedge q) \equiv (p \vee q) \wedge (\sim p \vee \sim q)$$

(b) Translate the statement $\forall x\,(C(x) \vee \exists y\,(C(y) \wedge F(x,y)))$ into English, where $C(x) : x$ has a computer and $F(x,y) : x$ and y are friends, and the domain for both x and y consists of all students in your school.

17. Determine the truth value of each of these statements, if the universe of discourse consists of all integers. In case of false value give a counter example.

    (i) $\exists n \left(n = -n^2\right)$

    (ii) $\forall n \left(n^3 \geq n\right)$

    (iii) $\exists n \, (n + 3 = 3n)$

    [Ans. (i) True, (ii) False, (iii) False]

18. (a) Using the vocabulary $Q(x) : x$ is a rational number, $R(x) : x$ is a real number, and $L(x) : x$ is less than 2 , translate the following into English sentences:

    (i) $\forall x \, (Q(x) \to R(x))$

    (ii) $\exists x \, (\neg Q(x) \wedge R(x))$

    (iii) $\forall x \, (Q(x) \wedge L(x) \to R(x) \wedge L(x))$

    (iv) $\forall x \, (Q(x) \wedge L(x)) \to \forall x \, (R(x) \wedge L(x))$.

    (b) Translate into logical expressions:

    (i) Each person in this room is either a BTech student or an MSc student.

    (ii) There is a student in this class room who speaks Hindi or English.

19. (a) Give the symbolic form of the following statements.

    (i) Some men are genius.

    (ii) For every x, there exists a y such that $x^2 + y^2 \geq 200$.

    (iii) Given any positive integer, there is a greater positive integer.

    (b) Test the validity of the following argument:

    If two sides of a triangle are equal, then the opposite angles are equal. Two sides of a triangle are not equal. Therefore, the opposite angles are not equal. [Ans. Invalid]

20. Write the following statements using predicates and quantifiers also find their negations

    (i) Some of the hardworking students do not get passed.

    (ii) No student completes homework but all get passed.

21. Let $A = \{1, 2, 3, 4, 5, 6\}$. Determine the truth value of each of the following:

    (i) $(\exists x \in A) \, (x^2 > 25)$

    (ii) $(\exists x \in A) \, (x + 6 > 12)$

    (iii) $(\forall x \in A) \, (x^2 - x < 30)$ [Ans. (i) True, (ii) False, (iii) False]

22. Let the universe of discourse be the set of real numbers and define

$$P(x, y) : \frac{x}{y} = 1.$$

    What is the truth value of the following statements? Justify your answers.

    (i) $\forall x \, \forall y \, P(x, y)$

(ii) $\forall x \exists y\, P(x, y)$

(iii) $\exists x \forall y\, P(x, y)$

(iv) $\exists x \exists y\, P(x, y)$ [Ans. (i) False, (ii) False, (iii) False, (iv) True]

23. Let $C(x, y)$ be the statement "x and y have chatted over the Internet," where the domain for the variables x and y consists of all students in your class. Use quantifiers to express each of these statements.

    (i) There are two students in your class who have not chatted with each other over the Internet.

    (ii) There is a student in your class who has chatted with everyone in your class over the Internet.

    (iii) There are at least two students in your class who have not chatted with the same person in your class.

    (iv) There are two students in the class who between them have chatted with everyone else in the class.

24. Let $C(x)$ be the statement "x has a cat," let $D(x)$ be the statement "x has a dog," and let $F(x)$ be the statement "x has a ferret." Express each of the following statements in terms of $C(x)$, $D(x)$, $F(x)$, quantifiers, and logical connectives. Moreover, write the negative of each of the statements. Let the domain consist of all students in your class.

    (i) A student in your class has a cat, a dog, and a ferret.

    (ii) All students in your class have a cat, a dog, or a ferret.

    (iii) Some student in your class has a cat and a ferret, but not a dog.

    (iv) No student in your class has a cat, a dog, and a ferret.

25. Let $P(x)$ be the statement "x can speak Russian" and let $Q(x)$ be the statement "x knows the computer language C++." Express each of these sentences in terms of $P(x)$, $Q(x)$, quantifiers, and logical connectives. The domain for quantifiers consists of all students at your school.

    (i) There is a student at your school who can speak Russian and who knows C++.

    (ii) There is a student at your school who can speak Russian but who doesn't know C++.

    (iii) Every student at your school either can speak Russian or knows C++.

    (iv) No student at your school can speak Russian or knows C++.

26. Use predicate calculus to prove the following agreement. "Students who pass the course do the homework or attend lectures. Bob did not attend every lecture. Bob passed the course. Therefore, Bob must have done the homework."

27. Verify the validity of the following argument. "Every living thing is a plant or an animal. John's goldfish is alive and it is not a plant. All animals have hearts. Therefore, John's goldfish has a heart." [Ans. Valid]

28. Show that the following argument is valid “One student in the class knows JAVA." “Everyone who knows JAVA can get a high-paying job." Therefore “Someone in this class can get a high-paying job."